

ECOLOGICAL SUCCESSION

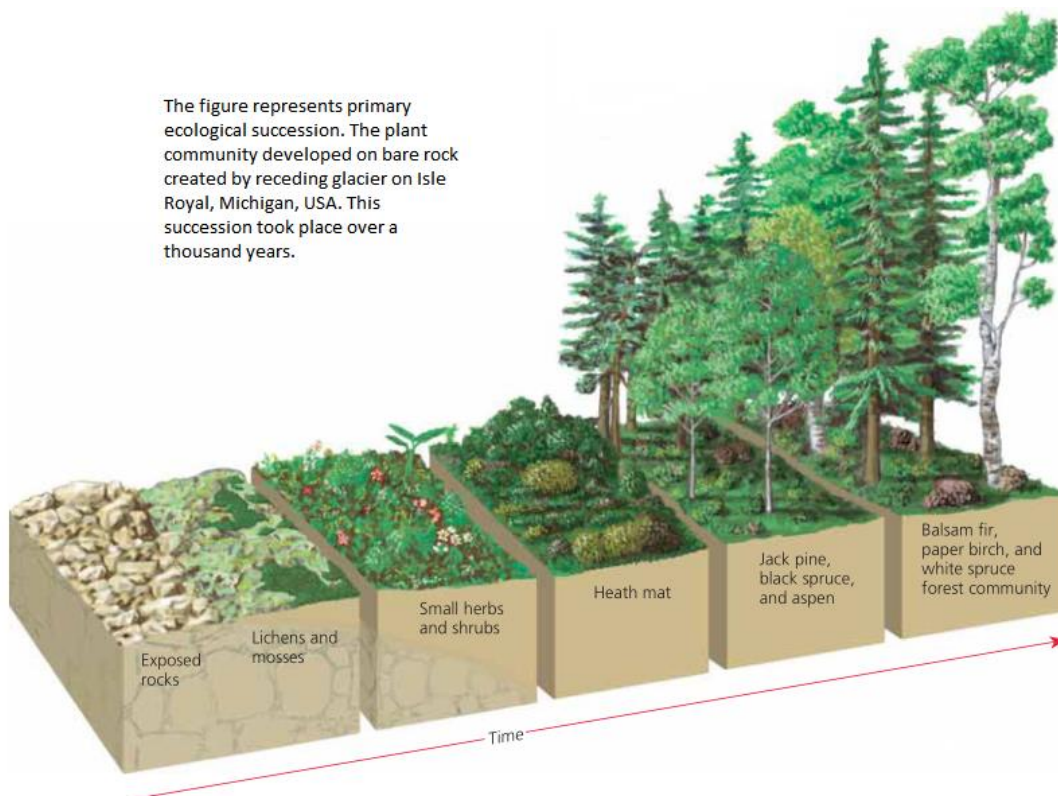
Gradual changes happening in species composition and processes of communities over time is known as ecological succession or community development. It is important to learn the process, rates, and pattern of ecological succession for the management of ecosystems.

Ecological succession can be divided into two major categories

- (i) Changes occurring over geological timescale (millions of years); also called paleo-ecological changes and
- (ii) Changes occurring over medium timescale (1 – 1000 years)

In another way, succession can also be classified into two types such as Primary and Secondary successions. Primary successions begins at a bare land where there is no life, whereas the secondary succession occurs at a place where the pre-existed ecosystem was either partially or fully destroyed by natural or unnatural means.

Primary Ecological Succession

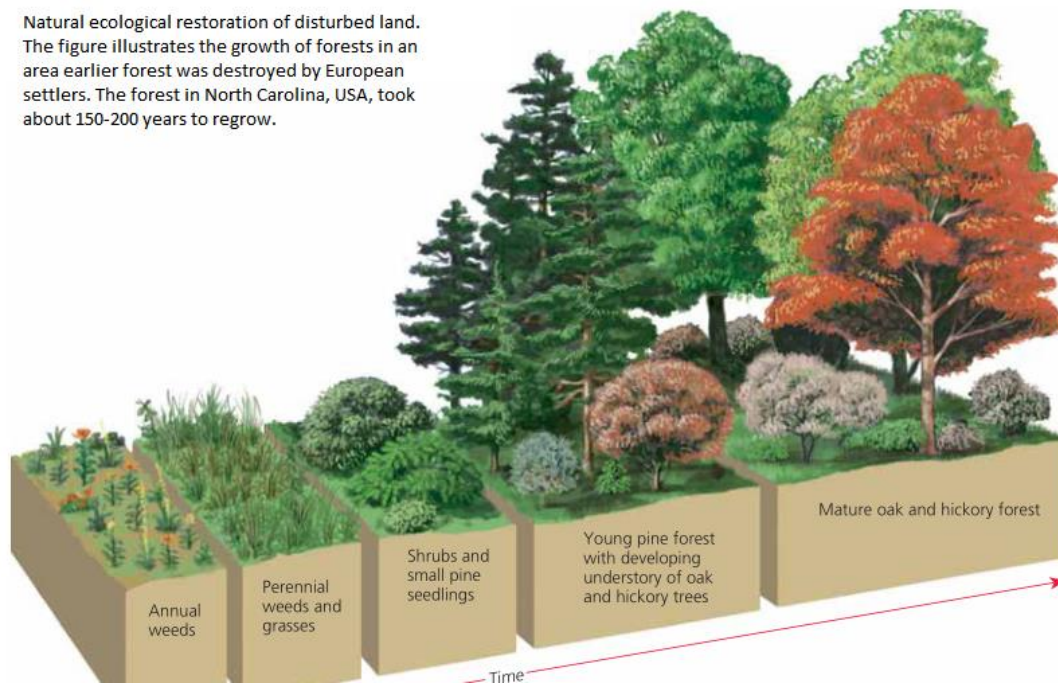


In primary succession, the slow process of soil formation begins with pioneer or early successional species, which arrive and attach themselves to inhospitable patches of the weathered rock. Lichens and mosses are examples of pioneer species. These species secrete mild chemicals and acids that penetrate the rocks and eventually make the soil fertile. Over time, the soil may be fertile and moist enough to support other plant species. Mid successional species such as herbs, grasses and low shrubs grow in the environment after lichens and mosses. Next, trees replace these shrubs over the next hundred to thousand years.

Secondary Ecological Succession:

In the secondary succession process, plant species grow in an area that has earlier been destroyed by natural wildfires, floods, or human intervention. Even though the surface species are destroyed, life remains under the soil, which eventually take foothold after the area is abandoned. We can consider the example of mature oak and pine forests of North Carolina, USA, which were destroyed by European settlers. They used the cleared forest for farming. As the nutrients of the land started to dwindle, the settlers moved on. The abandoned farmland underwent secondary succession as shown in the figure below.

Natural ecological restoration of disturbed land. The figure illustrates the growth of forests in an area earlier forest was destroyed by European settlers. The forest in North Carolina, USA, took about 150-200 years to regrow.



Secondary succession can take place 5-10 times faster than primary successions as most of the nutrients in the soil are still present, albeit at a much lower level than what is found in a live forest. The species that take hold as the leader at the end of the succession is called climax species.

The process of succession takes place via these steps:

1. **Nudation:** In this, the bare area is formed by one of several factors: volcanic eruption, landslide, flooding, fire, or other catastrophic event
2. **Invasion:** In this process, the arrival of an organism or many to the bare land takes place. These immigrant species are called "pioneers"
3. **Competition:** When the number of living organisms increases, the space, nutrients, etc. in the area are shared. The competition or struggle for existence can be intraspecific or interspecific. Because of the competition, the environment is modified unsuitably for the existing community, which is eventually replaced by the immigrant species. Co-action and co-existence is another seral that follows the competition seral.
4. **Climax:** The end process is called "climax" or "stabilization". When the climax community takes hold, the environment does not undergo further change unless by natural catastrophes or human intervention.

There are several other types of succession as well:

- a. Hydrosere – succession starting in a water environment
- b. Xerosere – succession starting in a dry, waterless environment
- c. Lithosere – succession starting in rocks
- d. Halosere – succession starting in a saline environment

Difference between xerosere and hydrosere

Xerosere (Dry)	Lithoseres	where the plants colonise bare rock	e.g. after glacial retreat, Snowdonia or a rocky shore, Oxwich Point, Gower or a newly created volcanic island (e.g. Surtsey, Iceland)
	Psammosere	where plants colonise coastal sand dunes	e.g. The south Gower - Oxwich Bay.
Hydrosere (Wet)	Hydroseres	where the plants colonise fresh water, as at a pond margin	e.g. Llanfihangel Gobion, Monmouthshire - ox-bow lakes
	Haloseres	where plants colonise salt marshes and sea estuaries	e.g. The North shore of the Gower, Llanrhidian salt marsh

Source courtesy: <https://qsstudy.com/difference-hydrosere-xerosere/>

Hydrosere

- Succession takes place in wet habitats
- Its first step is the submerged plant stage
- Plants of the first stage are- Elodia, Hydrilla, etc.
- Succession is limited to six steps such as- Submerged stage, floating stage, reed swamp stage, woodland stage, and climax forest stage.
- A hydrosere is a plant succession which occurs in an area of fresh water such as in oxbow lakes and kettle lakes.

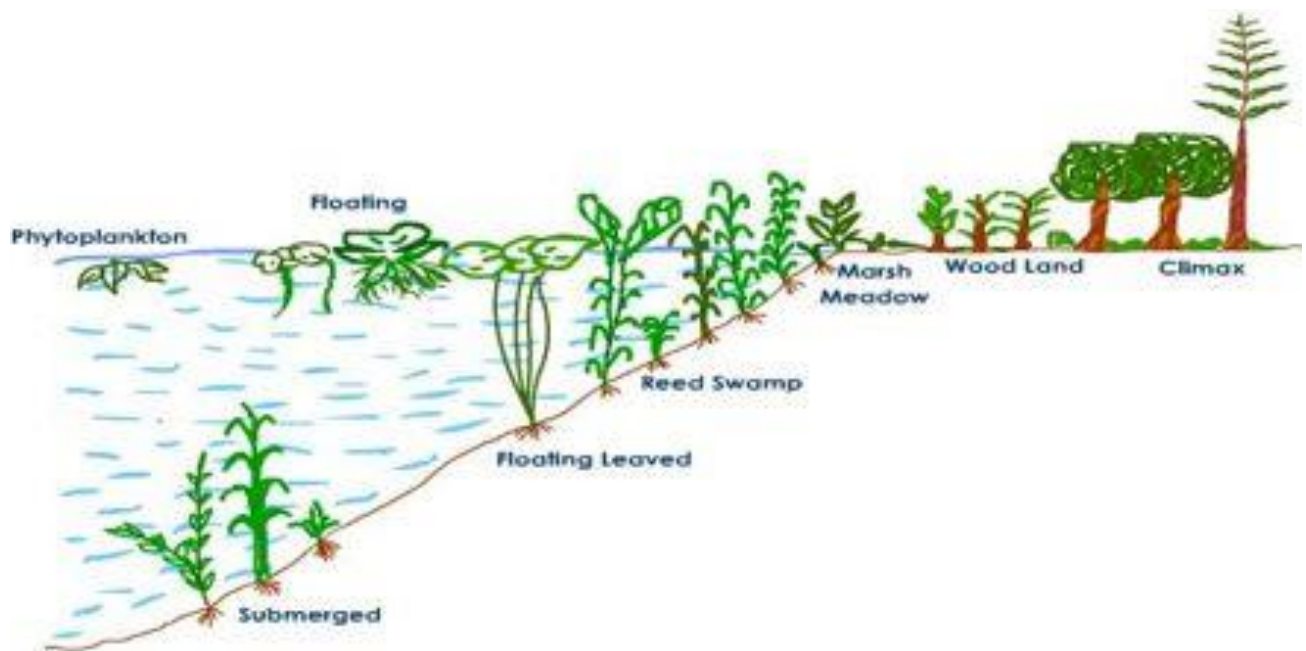


Fig: The different stage of Hydrosere (Source courtesy: <https://qsstudy.com/difference-hydrosere-xerosere/>)

Xerosere

- Succession begins with bare rocks, deserts; example: dry places.
- Its first step is the blue-green algae as well as thalloid lichen plant stage
- Plants of the first stage are Rhizocarpon, Rhinodina, etc.
- This succession is ended by six stages, such as – thalluid, licken stage, leafy licken stage, moss stage, herb stage, shrub stage, and climax forest Stage.
- The xerarch succession of ecological communities originated in the extremely dry situation such as sand deserts, sand dunes, salt deserts, rock deserts, etc.

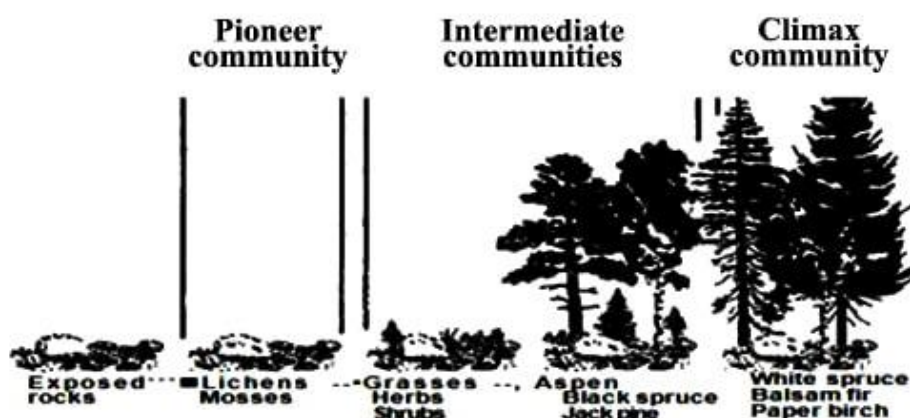


Fig: The different stage of Xerosere(Source courtesy: <https://qsstudy.com/difference-hydrosere-xerosere/>)

CLIMAX THEORY

1. MONOCLIMAX THEORY
 - This Theory was given by F.E. Clemens.
 - According to this theory, within a given region all land surface is eventually covered by a single type of community. This type of climax is determined by climate.
2. POLYCLIMAX THEORY
 - This theory was given by Tansley.
 - In this type, the climax vegetation does not consist of numerous type vegetation controlled by many factors .

FEATURES OF PIONEER SPECIES

The pioneer species should:

- have the habit of exploring new habitat.
- should be agile in nature.
- have greater ability to adapt to new environment.
- Have a wide choice of food.
- be a good breeder.
- be a tolerant species .

Summary:

The process of succession shows how a new community is established.

By the process of succession and by of the species of that area the factors responsible for succession can be determined.

The process of succession helps us in the conservation of the climax community.

Differences between primary and secondary succession

Primary succession	Secondary succession
Occurs in areas where there is no life and barren	Occurs in areas that were previously occupied, but devastated completely
Takes more than 1000 plus years	Takes just 50 to 100 years
No humus as no soil is seen in the initial step	Humus is present as there were previous occupants and decomposition took place between organisms that existed.
Goes through several seral communities	Less number of seral communities when compared to primary succession
Unfavourable starting point	Favourable and conducive environment
Eg: Bare rock, ponds, desert	Eg. The area affected by natural calamities, covered under deforestation